

Milan Jani

✉ milanjani707@gmail.com | ☎ +91 74859 75257 | 🌐 milanjani

🔗 milanjani | 📖 LeetCode | 📍 Rajkot



CAREER OBJECTIVE

Embedded systems undergraduate with hands-on experience in real-time control, automation, and safety-critical applications. Strong background in sensor-based systems, system-level debugging, and control-oriented software development, with a published patent in embedded IoT safety systems.

EDUCATION

Marwadi University, Bachelor of Technology in Information & Communication Technology (Pursuing) [July'23 – May'27]

- GPA: 9.03/10
- **Coursework:** Embedded Systems, Digital Electronics, Advanced Microprocessors, Computer Architecture, Programming Languages, Data Structures & Algorithms, Computer Networks

EXPERIENCE

VLSI Intern | *Microcircuits Innovations Pvt. Ltd., Ahmedabad* [June'25–July'25]

- Applied 50+ core Linux commands and Shell/TCL (Tool Command Language) scripting to automate repetitive design flow tasks using loops, conditionals, etc., improving overall lab process efficiency.
- Executed 10+ Makefile tasks and explored design tools to study circuit layout and backend flow.

Research Intern | *Information and Communication Technology, Marwadi University* [July'24-Jan'25]

- Conducted **research** on designing a **MIMO antenna** using the **finite element method (FEM)**, focusing on a low-profile, high-gain structure for **5G** and **satellite communication** through simulation.
- Evaluated **parameters** such as return loss, gain, and bandwidth to optimize antenna performance.

PUBLICATIONS

LPG Cylinder Monitoring and Leak Alert System | **Patent Publication** [June'24]

Patent No.: 202421033238 **Status:** *Published*

PROJECTS

Robotic Arm Development Project | *Project Contributor* | [Link](#) [Oct'25-Present]

- Applied robot control concepts using **NVIDIA Isaac Sim** and **Isaac Lab** for simulation and **RL training**.
- Contributed to robotic arm design and testing by integrating motion planning and control algorithms.

Adaptive Embedded Security and Automation System | [Link](#) [Oct'25]

- Developed **10+** embedded security **prototypes** including fire alarm, smart door lock, RFID entry, etc.
- Utilized **sensor modules** to improve **automation** efficiency and enable real-time threat detection.

Smart Vehicle Entry-Exit Logging System | [Link](#) [April'25]

- Built an automated gate logging solution using live number plate recognition, backend verification, and structured data storage, cutting manual entry work by **90%** and improving gate supervision.
- Proposed on-site deployment using a single-board controller to support card-tap-based visitor access, instant gate-pass slip generation, integrated printing, and smoother entry handling.

LPG Cylinder Monitoring and Leak Alert System

(Guides: Prof. C.D. Parmar, Prof. Vijay Dubey)

[Aug'23]

Course Project

- Built an IoT-based LPG system that tracks cylinder weight and sends alerts **2–3 days** early for booking.
- Implemented gas leak sensors and responsive LED alerts providing **24×7** real-time safety notifications.

SKILLS & TECHNOLOGIES

Languages	Microcontrollers	Tools & Platforms	Protocols
C, C++ , Embedded C, Python (Intermediate)	ARM, AVR, MSP430, Raspberry Pi, ESP32, ESP8266	Keil IDE, CCS, Arduino IDE, Proteus, Wokwi, Tinkercad, Blynk, ThingSpeak, Git, GitHub, Microwind	UART, SPI, I2C, CAN

Soft Skills: • Analytical thinking • Team collaboration • Adaptability • Time management

CERTIFICATIONS & PARTICIPATIONS

Digital Design using Verilog – NPTEL Online Certification Course

[Jan'26-Apr'26]

Semiconductor Fabrication and Manufacturing Overview Workshop

[Feb'26]

- Attended an industry-led workshop on semiconductor fabrication processes and manufacturing flow.

HOBBIES

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| • Embedded systems prototyping | • Technical reading | • Fitness training |
| • Problem Solving | • Playing volleyball | • Travelling |

REFERENCES

Prof. Chandrasinh Parmar

Head of Information & Communication Technology Department, Marwadi University

Email: chandrasinh.parmar@marwadieducation.edu.in

Prof. Dr. Sunil Lavadiya

Associate Professor, Information & Communication Technology, Marwadi University

Email: sunil.lavadiya@marwadieducation.edu.in